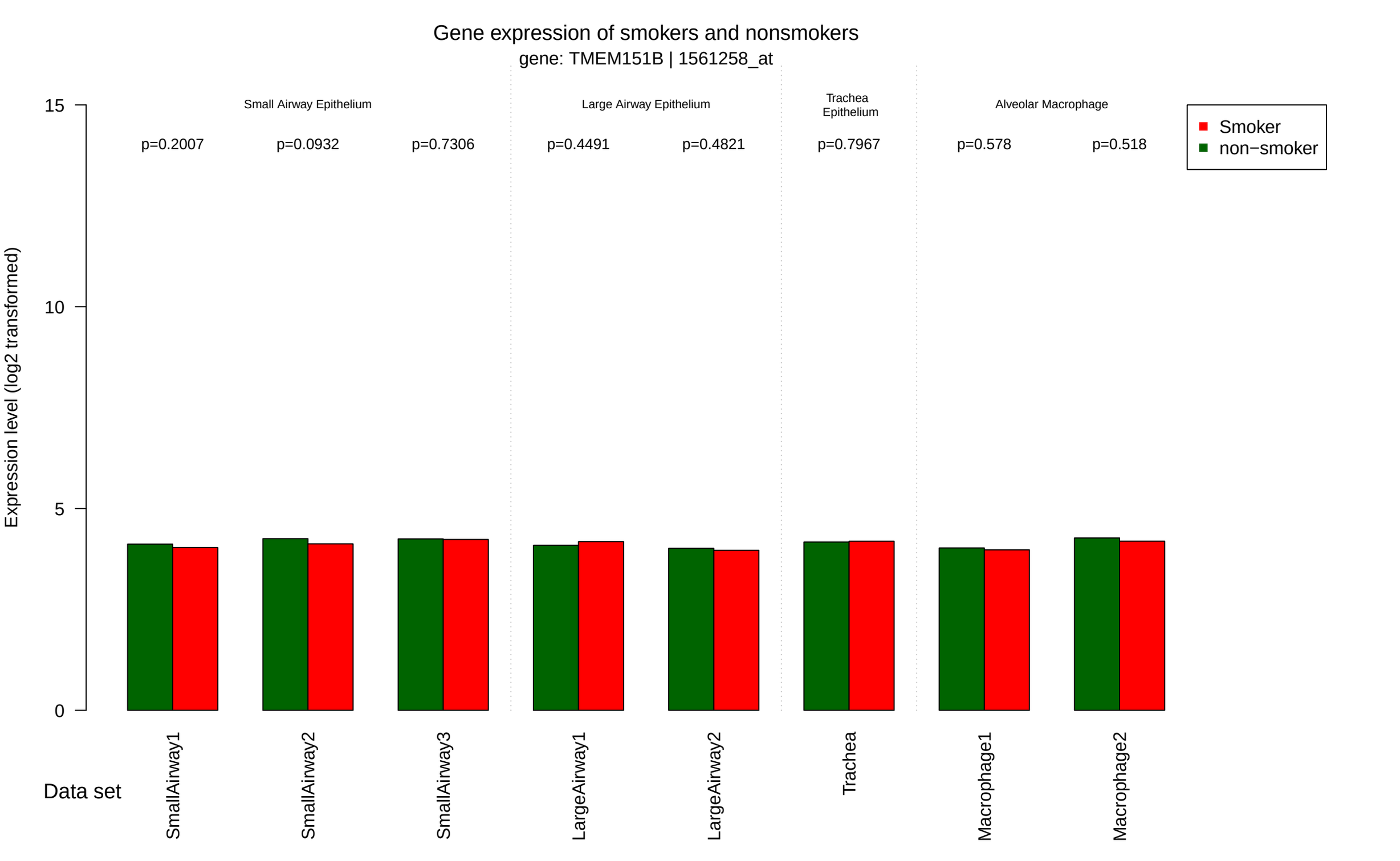
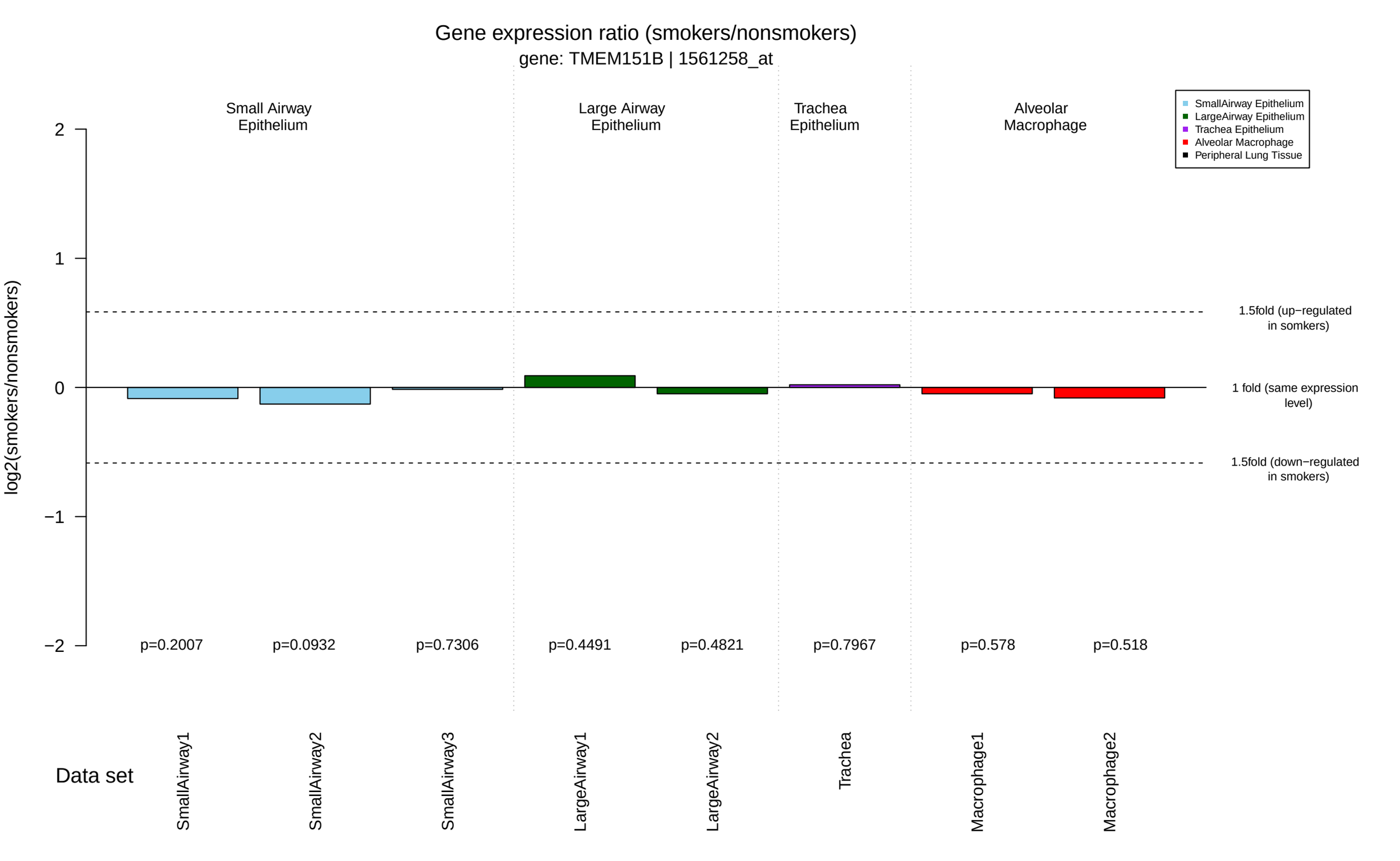
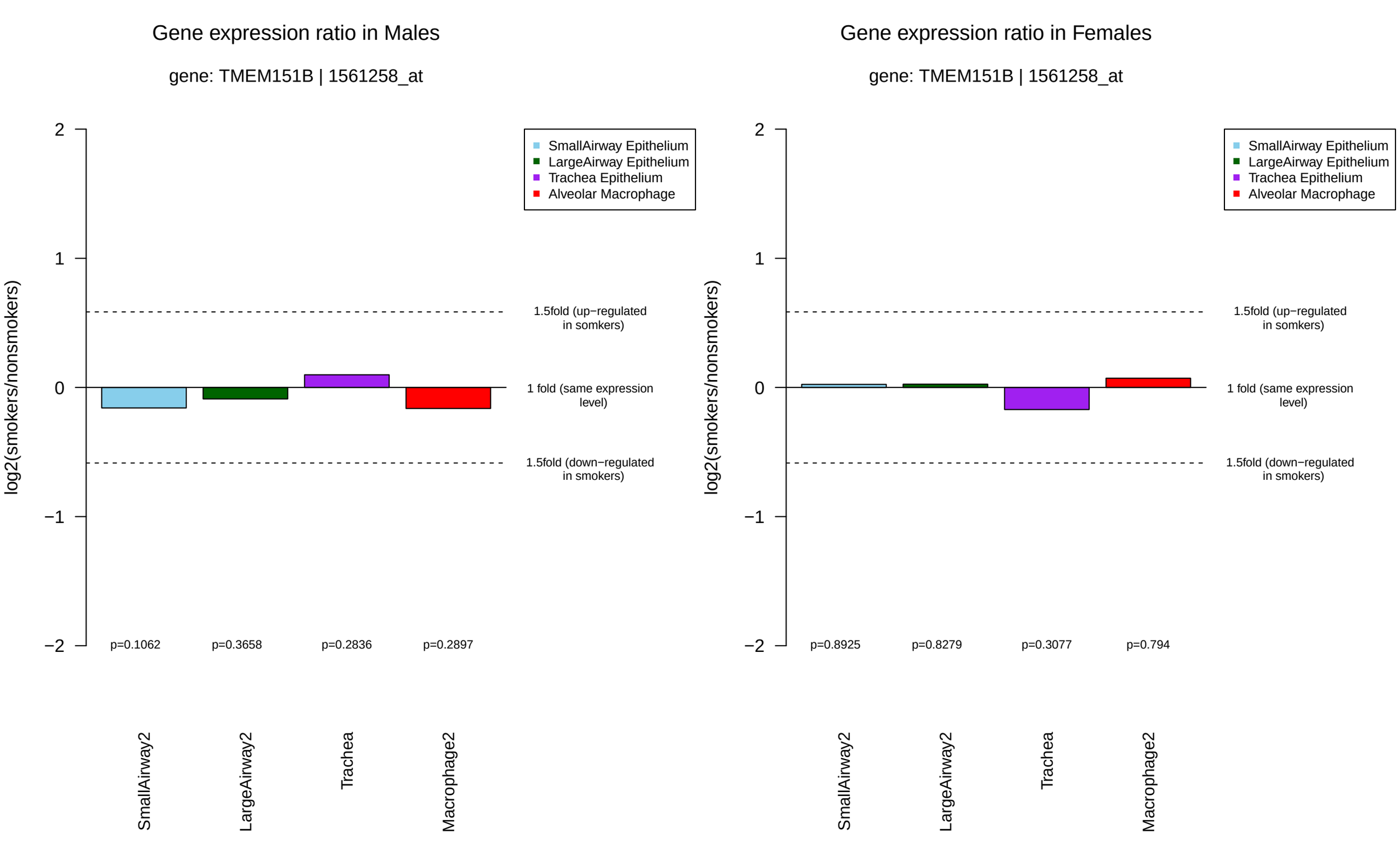
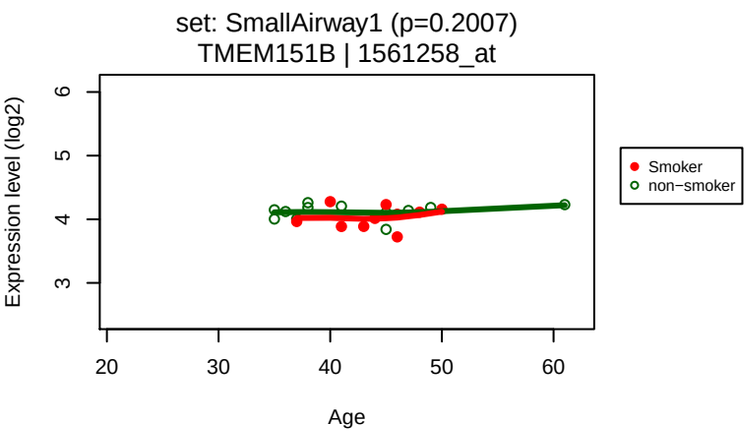
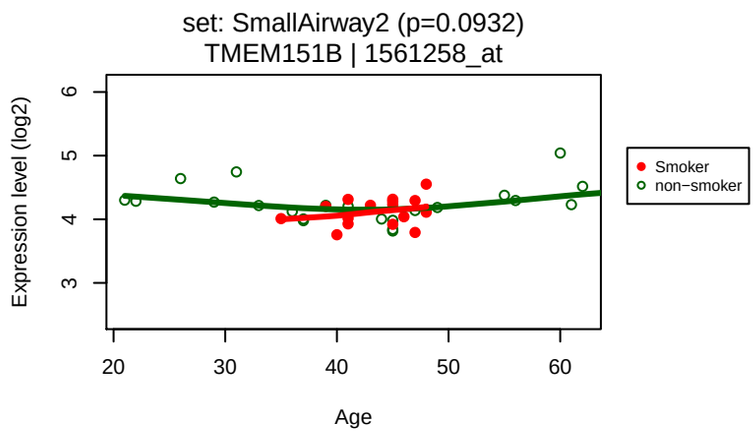
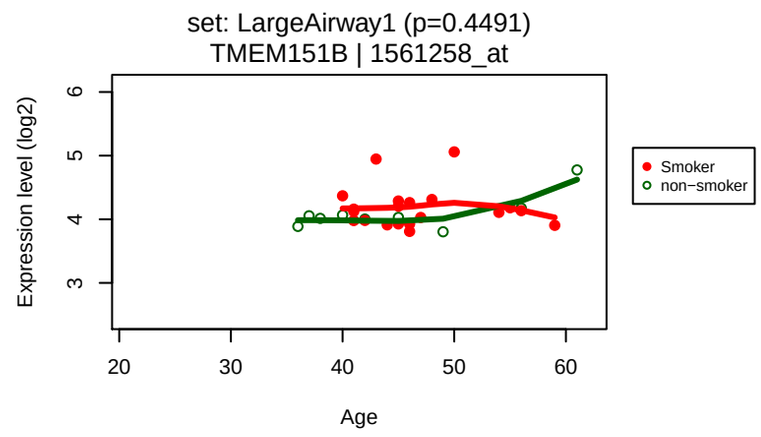

Gene expression of smokers and nonsmokers
gene: TMEM151B | 1561258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2007
p=0.0932
p=0.7306
p=0.4491
p=0.4821
p=0.7967
p=0.578
p=0.518
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

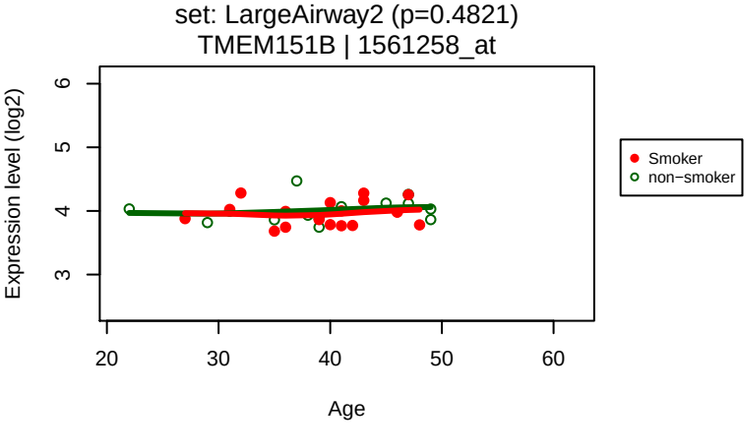

Gene expression ratio (smokers/nonsmokers)
gene: TMEM151B | 1561258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2007
p=0.0932
p=0.7306
p=0.4491
p=0.4821
p=0.7967
p=0.578
p=0.518
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

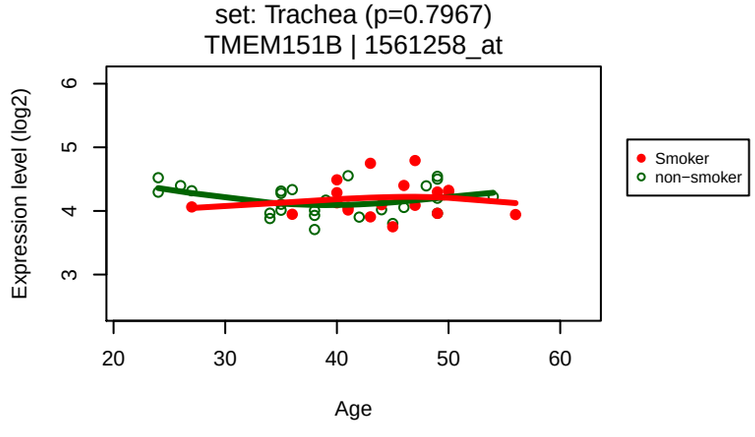

Gene expression ratio in Males
gene: TMEM151B | 1561258_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1062
p=0.3658
p=0.2836
p=0.2897
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM151B | 1561258_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8925
p=0.8279
p=0.3077
p=0.794
SmallAirway2
LargeAirway2
Trachea
Macrophage2

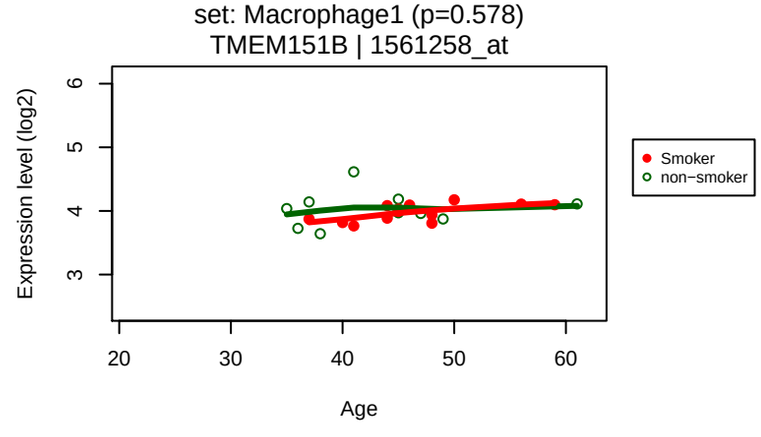

set: SmallAirway1 (p=0.2007)
TMEM151B | 1561258_at
Expression level (log2)
Age
Smoker
non−smoker

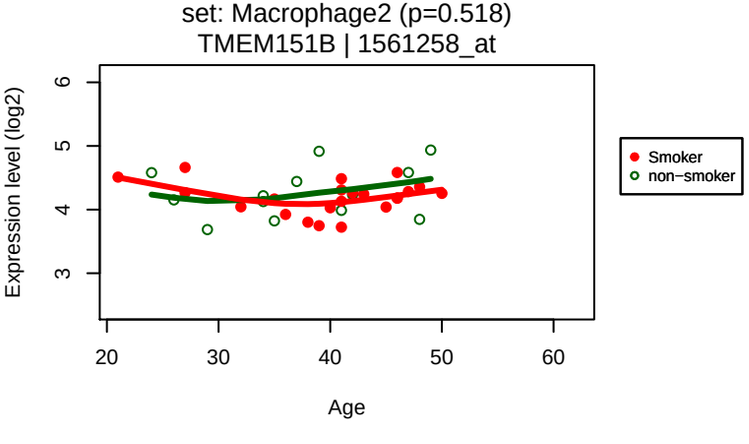

set: SmallAirway2 (p=0.0932)
TMEM151B | 1561258_at
Expression level (log2)
Age
Smoker
non−smoker

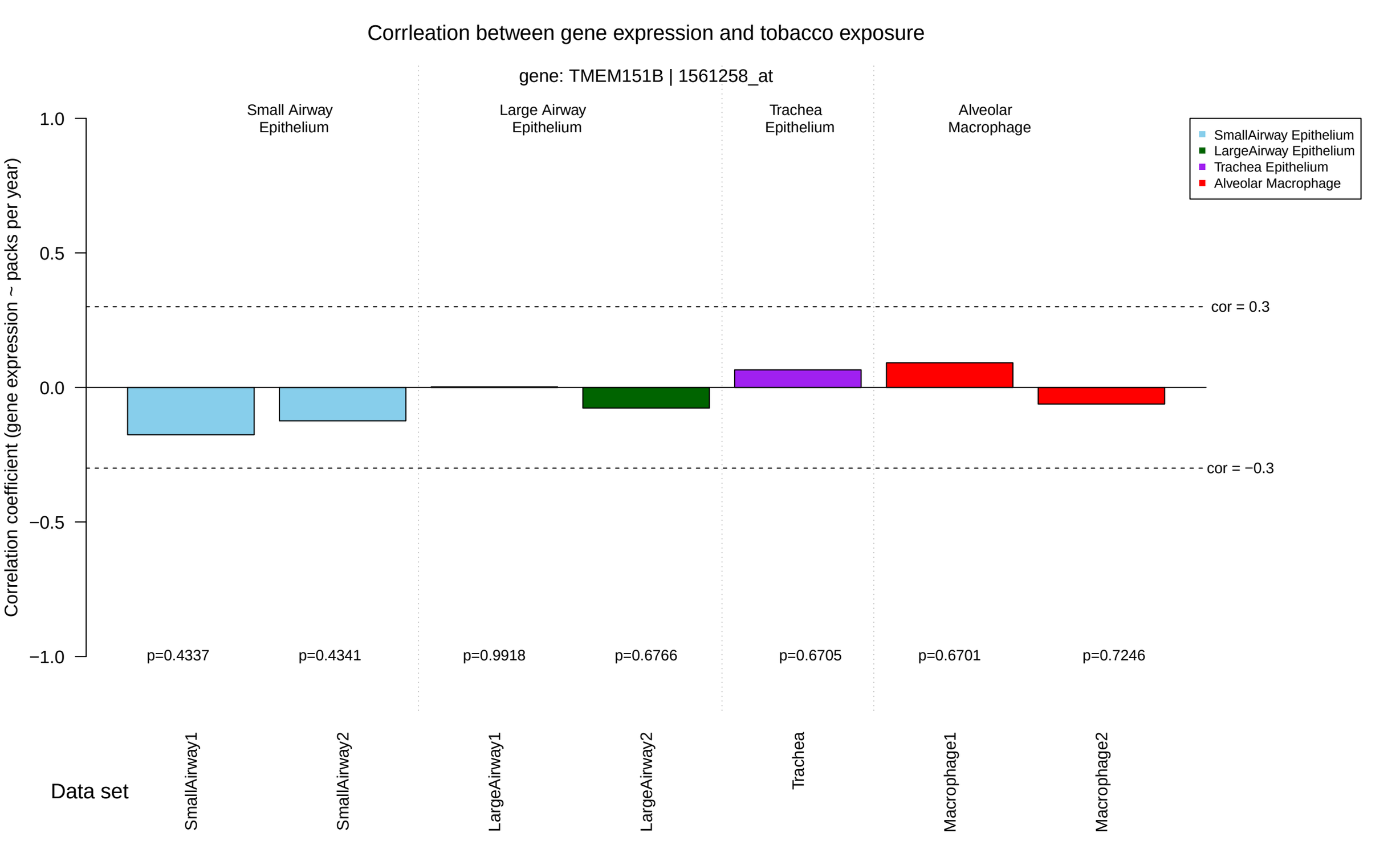

set: LargeAirway1 (p=0.4491)
TMEM151B | 1561258_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4821)
TMEM151B | 1561258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7967)
TMEM151B | 1561258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.578)
TMEM151B | 1561258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.518)
TMEM151B | 1561258_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM151B | 1561258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4337
p=0.4341
p=0.9918
p=0.6766
p=0.6705
p=0.6701
p=0.7246
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2